Solutions for Homework 3

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1.

```r
# Load data
df <- read.table("data/HW3-1.txt", header = TRUE)
X <- df[, -1]        # Extract covariates
```

(a)

```r
(x.bar <- colMeans(X))
```

```
##   Sales  Profit  Assets
## 155.603  14.704 710.911
```

```r
(S <- cov(X))
```

```
##              Sales      Profit     Assets
## Sales    7476.4532   303.61862 -35575.960
## Profit    303.6186    26.19032  -1053.827
## Assets -35575.9596 -1053.82739 237054.270
```

(b)

```r
# Extract (Sales, Profit) data
X.SP <- X[, c("Sales", "Profit")]
x.bar.SP <- colMeans(X.SP)
S.SP <- cov(X.SP)
S.SP.inv <- solve(S.SP)

# Compute squared generalized distances
d2.SP <- apply(X.SP, 1, function(x) {
    t(x - x.bar.SP) %*% S.SP.inv %*% (x - x.bar.SP)
})

# Compute proportion of data falling within cutoff
chi2.median <- qchisq(0.5, 2)
mean(d2.SP <= chi2.median)
```

```
## [1] 0.4
```

This means that 40% of the data (based on sales, profit) falls within the 50% contour.

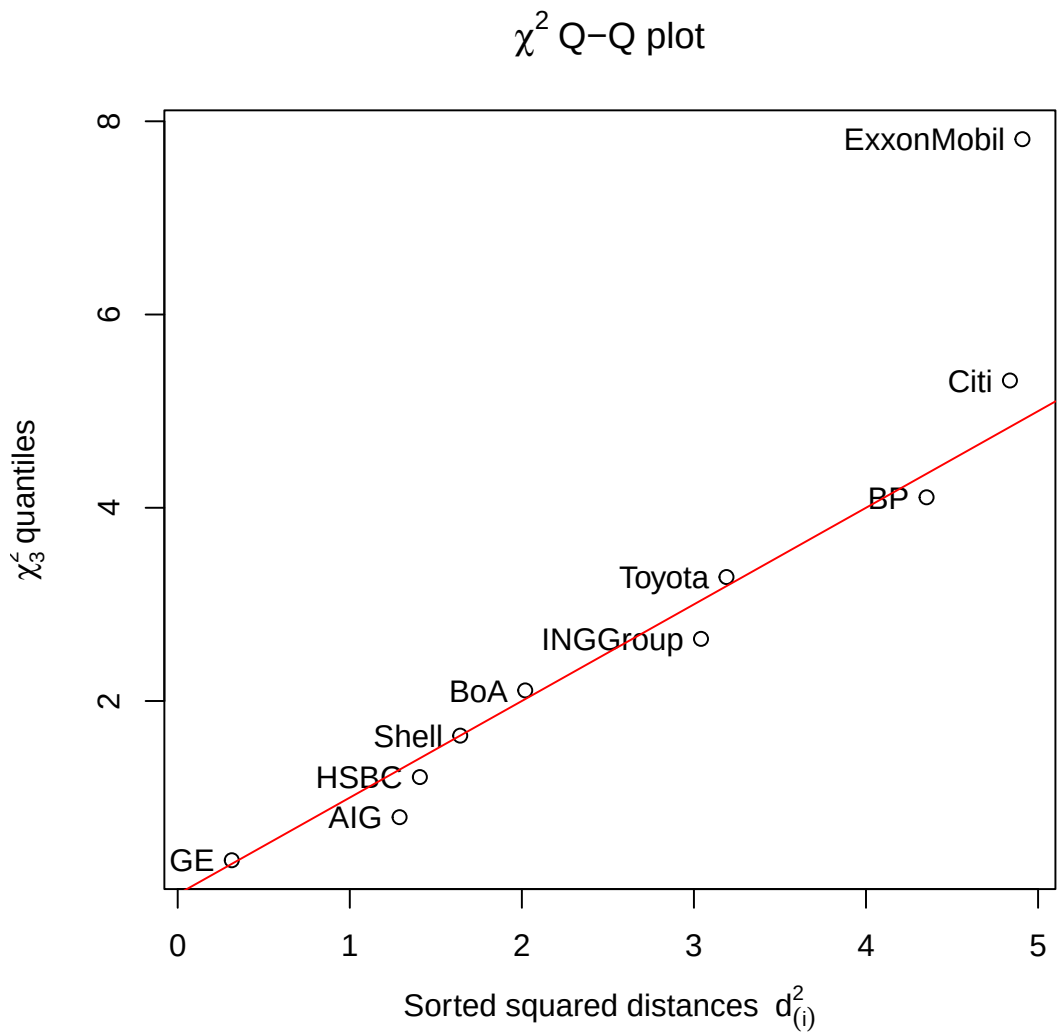
(c)

```r
# Compute squared generalized distances
S.inv <- solve(S)
d2 <- apply(X, 1, function(x) {
    t(x - x.bar) %*% S.inv %*% (x - x.bar)
})

n <- nrow(X)
p <- ncol(X)
```

```r
chi2.q <- qchisq((1:n - 0.5)/n, df = p)     # Chi-squared quantiles

# Plot chi-squared quantiles against d2
plot(
    sort(d2), chi2.q,
    xlim = c(min(d2) - 0.2, max(d2)),
    xlab = expression("Sorted squared distances " ~ d[(i)]^2),
    ylab = expression(chi[3]^2 ~ "quantiles"),
    main = expression(chi^2 ~ "Q-Q plot")
)
text(
    sort(d2) - 0.1, chi2.q, df[order(d2), "Company"],
    adj = c(1, 0.5)
)
abline(0, 1, col = "red")   # Reference line (perfect alignment)
```



The data for ExxonMobil lies quite far away from the red reference line, suggesting that it may

be an outlier, and indicating a possible violation of normality.

2. (a) The training SSE for model 2 will *always* be lower than (or equal to) that of model 1, because model 1 is contained within model 2.

Remark: We have model containment in the sense that restricting the parameter $\boldsymbol{\beta}$ from $\mathbb{R}^4$ in model 2 to $\mathbb{R}^2 \times \{0\}^2$ yields model 1. Thus,

$$\begin{aligned}
\mathrm{SSE}_{1,\mathrm{train}} &:= \inf_{\boldsymbol{\beta}\in\mathbb{R}^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 \\
&= \inf_{\boldsymbol{\beta}\in\mathbb{R}^2\times\{0\}^2} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2 - \beta_3 x_i^3)^2 \\
&\geq \inf_{\boldsymbol{\beta}\in\mathbb{R}^4} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2 - \beta_3 x_i^3)^2 \\
&=: \mathrm{SSE}_{2,\mathrm{train}},
\end{aligned}$$

where the sums are over training data.

(b) Since the true data generating process is linear (aligning with model 1), model 2 will be more prone to overfitting the training data, and is likely to suffer greater test SSE than model 1 on typical test data.

(c)
```r
n.simulations <- 1000
n.train <- 100
n.test  <- 100

# True parameters
beta.0 <- 5
beta.1 <- 2
sigma  <- 1

SSE.diffs <- replicate(n.simulations, {
    # Generate training data
    X.train  <- rnorm(n.train, 0, 5)
    Y.train  <- beta.0 + beta.1 * X.train + rnorm(n.train, 0, sigma)
    df.train <- data.frame(X = X.train, Y = Y.train)

    # Fit models
    model.linear <- lm(Y ~ X, data = df.train)
    model.cubic  <- lm(Y ~ X + I(X^2) + I(X^3), data = df.train)

    # Compute train SSEs
    SSE.linear.train <- sum(residuals(model.linear)^2)
    SSE.cubic.train  <- sum(residuals(model.cubic)^2)

    # Generate testing data
    X.test  <- rnorm(n.test, 0, 5)
    Y.test  <- beta.0 + beta.1 * X.test + rnorm(n.test, 0, sigma)
    df.test <- data.frame(X = X.test, Y = Y.test)

    # Compute test SSEs
    SSE.linear.test <- sum((Y.test - predict(model.linear, newdata = df.test))^2)
    SSE.cubic.test  <- sum((Y.test - predict(model.cubic, newdata = df.test))^2)

    # Keep SSE differences (cubic - linear)
    c(
```

```
        SSE.cubic.train - SSE.linear.train,
        SSE.cubic.test  - SSE.linear.test
    )
})

SSE.diffs <- as.data.frame(t(SSE.diffs))
colnames(SSE.diffs) <- c("Train", "Test")
summary(SSE.diffs)  # SSE.cubic - SSE.linear

##      Train                 Test
##  Min.   :-1.461e+01   Min.   : -9.98816
##  1st Qu.:-2.727e+00   1st Qu.: -0.03427
##  Median :-1.387e+00   Median :  1.61874
##  Mean   :-2.008e+00   Mean   :  4.17529
##  3rd Qu.:-6.169e-01   3rd Qu.:  4.73028
##  Max.   :-8.715e-04   Max.   :301.68417

mean(SSE.diffs[, "Test"] > 0)

## [1] 0.74
```

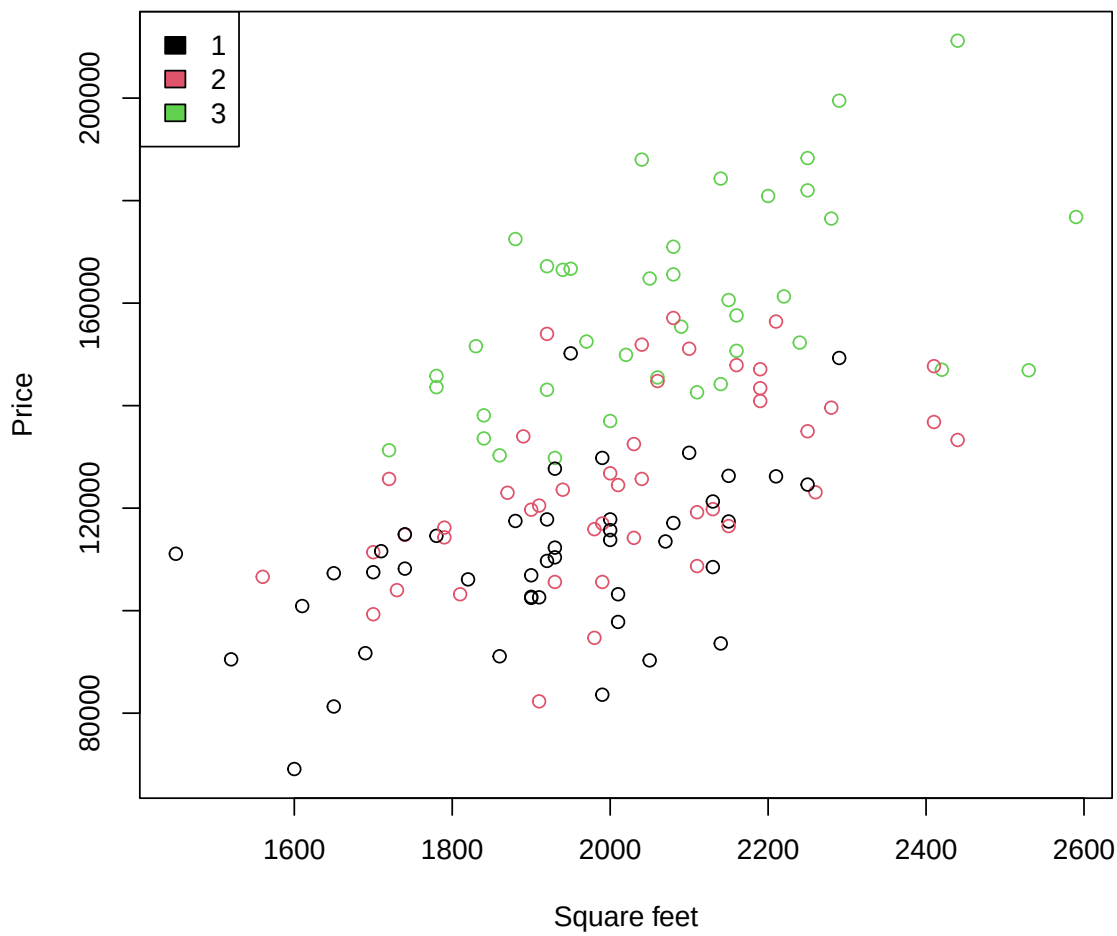
Note that the training SSE differences (cubic – linear) are always negative, as predicted in part (a). The test SSE for the cubic model is higher than that of the linear model more often than not (around 74% of the time).

3.

```r
# Load data
df <- read.table("data/HW3-house.csv", sep = ",", header = TRUE)
df$nbhd <- as.factor(df$nbhd)
summary(df)
```

```
##       home            nbhd         sqft          price
##  Min.   :  1.00   nbhd01:44   Min.   :1450   Min.   : 69100
##  1st Qu.: 32.75   nbhd02:45   1st Qu.:1880   1st Qu.:111325
##  Median : 64.50   nbhd03:39   Median :2000   Median :125950
##  Mean   : 64.50               Mean   :2001   Mean   :130427
##  3rd Qu.: 96.25               3rd Qu.:2140   3rd Qu.:148250
##  Max.   :128.00               Max.   :2590   Max.   :211200
```

```r
plot(
    df$sqft, df$price, col = df$nbhd,
    xlab = "Square feet",
    ylab = "Price"
)
legend("topleft", legend = paste(1:3), fill = 1:3)
```



(a)
```
model.simple <- lm(price ~ sqft, data = df)
coef(model.simple)
```
```
##  (Intercept)         sqft
## -10091.12991     70.22632
```

This suggests that an extra square foot adds an average of \$70.226 to the value of a house.

(b)
```
n <- nrow(df)
p <- 1
X <- model.matrix(model.simple)          # Design matrix
H <- X %*% solve(t(X) %*% X) %*% t(X)    # Hat matrix
df[diag(H) > 2 * (p + 1) / n, "home"]    # Indices of high-leverage data points
```
```
##  [1]  15  29  38  41  62  66  84  85  97 103 104 106
```

(c)
```
model.parallel <- lm(price ~ sqft + nbhd, data = df)
coef(model.parallel)
```
```
## (Intercept)        sqft  nbhdnbhd02  nbhdnbhd03
## 21241.17443    46.38592 10568.69781 41535.30643
```

This suggests that an extra square foot adds an average of \$46.386 to the value of a house.

(d) Neighborhood is a confounding variable, which explains variation in house prices (different neighborhoods have different distributions of house prices), but is not accounted for in the simple model in part (a). The parallel lines model in part (c) controls for this variable, and lets us isolate the effect of square footage.

(e)
```
model.full <- lm(price ~ sqft * nbhd, data = df)
beta <- coef(model.full)
beta["sqft"] + 0
```
```
##     sqft
## 40.30018
```
```
beta["sqft"] + beta["sqft:nbhdnbhd02"]
```
```
##    sqft
## 49.4285
```
```
beta["sqft"] + beta["sqft:nbhdnbhd03"]
```
```
##     sqft
## 49.32586
```

This suggests that an extra square foot adds averages of \$40.300, \$49.429, \$49.326 to the value of a house in neighborhoods 1, 2, 3 respectively.

(f)
```
anova(model.parallel, model.full)
```
```
## Analysis of Variance Table
##
## Model 1: price ~ sqft + nbhd
## Model 2: price ~ sqft * nbhd
##   Res.Df        RSS Df Sum of Sq      F Pr(>F)
## 1    124 2.8876e+10
## 2    122 2.8783e+10  2  93228968 0.1976  0.821
```

Note that the F-test (comparing SSEs between the two models) yields a p-value of 0.821. This suggests that the full model in part (e) does not significantly explain the variation in house prices

better than the parallel lines model in part (c).

Remark: This test uses the statistic

$$F := \frac{(\mathrm{SSE}_1 - \mathrm{SSE}_2)/(\nu_1 - \nu_2)}{\mathrm{SSE}_2/\nu_2}$$

where ν_i are the degrees of freedom $n - p_i$ of the models, with the null distribution $F_{\nu_1-\nu_2,\nu_2}$.

(g) We conclude that the parallel lines model in part (c) is sufficient to model the variation of house prices, and answer our research question: an extra square foot adds an average of \$46.386 to the value of a house.

We trust the parallel lines model (c) over the simple model (a) since the latter does not control for neighborhoods at all and gives a biased estimate of the effect of square footage. We have also determined that the parallel lines model is not significantly worse than the full interaction model in part (e).

4. (a) The likelihood function for this model is

$$\mathcal{L}(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^{n} f(y_i; x_i, \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2\right).$$

Plugging in the MLEs, we only need

$$\hat{\sigma}^2_{\text{MLE}} = \frac{\text{SSE}}{n} := \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_{0,\text{MLE}} - \hat{\beta}_{1,\text{MLE}} x_i)^2$$

to obtain

$$\sup_{(\beta_0,\beta_1,\sigma^2)\in\mathbb{R}\times\mathbb{R}\times\mathbb{R}_+} \mathcal{L}(\beta_0, \beta_1, \sigma^2) = \mathcal{L}(\hat{\beta}_{0,\text{MLE}}, \hat{\beta}_{1,\text{MLE}}, \hat{\sigma}^2_{\text{MLE}}) = \left(\frac{2\pi}{n}\,\text{SSE}\right)^{-n/2} \exp\left(-\frac{n}{2}\right).$$

Next, when $\beta_1 = 0$,

$$\mathcal{L}(\beta_0, 0, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0)^2\right),$$

which is identical to the likelihood of the model $y_i \overset{iid}{\sim} \mathcal{N}(\beta_0, \sigma^2)$. Recalling the corresponding MLEs, we conclude that $\mathcal{L}(\beta_0, 0, \sigma^2)$ attains its maximum at

$$\hat{\beta}_0 = \bar{y}, \qquad \hat{\sigma}^2 = \frac{\text{SST}}{n} := \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2,$$

so

$$\sup_{(\beta_0,\beta_1,\sigma^2)\in\mathbb{R}\times\{0\}\times\mathbb{R}_+} \mathcal{L}(\beta_0, \beta_1, \sigma^2) = \left(\frac{2\pi}{n}\,\text{SST}\right)^{-n/2} \exp\left(-\frac{n}{2}\right).$$

Thus, the likelihood ratio statistic for testing

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_a : \beta_1 \neq 0$$

is

$$\Lambda(\boldsymbol{Y}) := \frac{\sup_{\mathbb{R}\times\{0\}\times\mathbb{R}_+} \mathcal{L}}{\sup_{\mathbb{R}\times\mathbb{R}\times\mathbb{R}_+} \mathcal{L}} = \left(\frac{\text{SST}}{\text{SSE}}\right)^{-n/2} = \left(\frac{\text{SSE}}{\text{SST}}\right)^{n/2}.$$

(b) Using SST = SSE + SSR, where

$$\text{SSR} := \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 := \sum_{i=1}^{n}(\hat{\beta}_{0,\text{MLE}} + \hat{\beta}_{1,\text{MLE}} x_i - \bar{y})^2,$$

the rejection region $\{\boldsymbol{Y} : \Lambda(\boldsymbol{Y}) < k\} = \{(\text{SSE}/\text{SST})^{n/2} < k\}$ can be reformulated as

$$\left\{\frac{\text{SST}}{\text{SSE}} > k^{-2/n}\right\} = \left\{1 + \frac{\text{SSR}}{\text{SSE}} > k^{-2/n}\right\} = \left\{F := \frac{\text{SSR}}{\text{SSE}/(n-2)} > c := (n-2)(k^{-2/n} - 1)\right\}.$$

Remark: We have assumed that $n > 2$.

(c) Given that $F \sim F_{1,n-2}$ under H_0, the critical value c for a level-α test must satisfy $\mathbb{P}_{H_0}(F > c) = \alpha$, hence we set $c = F_{1,n-2;\alpha}$ to be the upper-α quantile of the $F_{1,n-2}$ distribution.

Putting $\alpha = 0.01$, $n = 120$, the critical value is

```r
n <- 120
qf(0.01, 1, n - 2, lower.tail = FALSE)

## [1] 6.854641
```

When SSE $= 45.818$ and SSR $= 3.58$, the F-statistic can be computed as

```r
SSE <- 45.818
SSR <-  3.58
(F.stat <- SSR/(SSE/(n - 2)))

## [1] 9.219957
```

which exceeds the previously computed critical value. Thus, we reject the null hypothesis.

Remark: Note that the given data yields a p-value of

```r
1 - pf(F.stat, 1, n - 2)

## [1] 0.002946696
```

which is less than the level of significance $\alpha = 0.01$ for this test.

(d) The likelihood ratio statistic obeys $-2\log\Lambda(\boldsymbol{Y}) \xrightarrow{d} \chi_1^2$ as $n \to \infty$ under H_0, via Wilk's Theorem. Approximating the null distribution of

$$-2\log\Lambda = n\log\left(\frac{\text{SST}}{\text{SSE}}\right) = n\log(\text{SSE}+\text{SSR}) - n\log(\text{SSE})$$

by χ_1^2, the test statistic

```r
(D <- n * log(SSE + SSR) - n * log(SSE))

## [1] 9.027949
```

is greater than the critical value

```r
qchisq(0.01, 1, lower.tail = FALSE)

## [1] 6.634897
```

Thus, we reject the null hypothesis.

Remark: Note that the rejection region $\{\Lambda < k\}$ is equivalent to $\{-2\log\Lambda > c'\}$ for the critical value c'.

The corresponding p-value is

```r
1 - pchisq(D, 1)

## [1] 0.002658826
```

which is very close to that obtained by the exact F-test.